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Fitting innovations into existing categories: Evidence from mortgage-backed securities

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Abstract

Research Summary: What role do innovators play in the evolution of product categories? While management scholars have documented multiple cases of innovators creating new product categories, less attention has been paid to how innovators reshape existing ones. This paper explores the possibility that innovators can change the boundaries of established categories to accommodate their products. In an historical case study of the acceptance of mortgage-backed securities (MBS) as bonds in the United States between 1970 and 1995, I find that MBS issuers succeeded in fitting their products into the bond category by introducing a new dimension to the category.

Managerial Summary: This study expands our understanding of category evolution by showing that innovators can reshape the boundaries of existing product categories to accommodate new offerings. Specifically, it finds that innovators succeeded in modifying a prototype-based category by positioning their product as a substitute—a goal-based category. Redefining category boundaries allowed the innovators to secure membership for their products without needing to create a new category or fully redesign the product. This research highlights the agentic role of innovators in the evolution of categories, the interaction between

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goal-based and prototype-based categories in innovation diffusion, and the importance of commensuration for achieving product-category fit.

KEYWORDS

category evolution, commensuration, goal-based categories, historical methods, innovation

1 | INTRODUCTION

In positioning their products within categories, innovators face an important choice: create a new category or fit their products into an existing one (e.g., Anthony et al., 2016; Pontikes, 2018). While several studies have examined how innovators create new product categories (e.g., Boghossian & David, 2021; Khaire & Wadhvani, 2010), the process by which they fit products into existing categories has received comparatively less attention. This oversight is surprising given the critical role that category membership plays in extending the appeal of innovation beyond early adopters to mainstream customers (e.g., Moore, 2014), as well as the costly and uncertain nature of new category creation (Rhee et al., 2017). Understanding how innovators fit their products into existing categories is essential for explaining firms' positioning strategies, the diffusion of innovations, and the evolution of product categories.

This lack of research on how innovators fit new products into existing categories is partly due to the assumptions that fitting innovations into existing categories is relatively straightforward and that any difficulties in the process can be addressed through product (re)design (e.g., Hannan et al., 2012; Zuckerman, 2000). However, we know little about the conditions under which (re)design can effectively secure category membership, or what strategies are available when such efforts fall short.

Drawing on an in-depth historical case study of the acceptance of mortgage-backed securities (MBS) as bonds in the United States, I examine the process by which innovators fit their products into an existing category. Taking a longitudinal perspective, I explore the multi-decade evolution of both the innovation's design and the customers' evaluation criteria. I find that, in promoting the acceptance of MBS as bonds—a preexisting category of financial products—innovators went beyond redesigning the product. Specifically, they succeeded by adding a new dimension to the existing category: prepayment risk. This process involved transforming the uncertainty about mortgage prepayments—a key difference between mortgages and bonds—into a type of measurable risk, and then aligning that risk with call risk (the risk of early bond repayment). My findings suggest that altering the boundaries of an existing category can complement product design in facilitating category fit and, consequently, the diffusion of innovation.

This investigation makes four contributions. First, it highlights the interplay between product (re)design and category boundary changes in innovation diffusion. Specifically, the findings suggest that fitting a new product into an existing category can involve not just redesigning the product but also modifying the category itself. Second, by demonstrating how innovators reshape category boundaries, I identify a broader range of strategies firms can use for product positioning. Third, by analyzing how innovators make their products commensurate with existing categories, this study underscores the importance of commensuration in how customers



evaluate innovation. Finally, this work connects the demand-based view of strategy (e.g., Adner, 2002; Adner & Zemsky, 2006; Priem, 2007) with research on category strategy and innovation.

2 | CATEGORIES AND INNOVATION

A product category is “a shared meaning between the actors [who are] party to an economic exchange about that which is being exchanged” and “whose boundaries are defined by common features” (David & Lee, 2022, p. 717). As this definition suggests, categories consist of sets of products characterized by a label and a set of rules for including products into that category (e.g., Navis & Glynn, 2011). For example, the category “umbrella” may include a representative set of products, the category label (“umbrella”), and inclusion criteria such as “a collapsible shade for protection against weather consisting of fabric stretched over hinged ribs radiating from a central pole” (Merriam-Webster.com). A product category can be characterized by a test code—a minimum set of criteria a product must meet to be considered a member (e.g., Hannan et al., 2012).

The idea that categorization facilitates the diffusion of innovation stems from studies that conceptualize the buyer decision-making as a two-step process (e.g., Ratneshwar & Shocker, 1991; Rosa et al., 1999; Urban et al., 1993). The first step involves eliminating alternatives that are less relevant to the buyer's search for a solution. The second step involves selecting the best option from the resulting consideration set. Within this framework, an innovation must belong to the desired category for it to be included in the customer's consideration set (Ratneshwar & Shocker, 1991).

If an innovation fails to meet a category's test code, it may be excluded from the consideration set. This exclusion may be due to customers' inability to recognize the innovation as relevant to their needs. For example, a product that does not offer a collapsible shade is unlikely to be recognized or accepted as an umbrella. Consequently, firms commercializing innovations that do not meet the test code of an existing category may face penalties in the form of reduced demand and lower prices; customers often reject products they do not recognize as category members (e.g., Hsu, 2006; Hsu et al., 2009). Categorization may be particularly important for social goods—such as financial securities—whose value “to a given consumer depends heavily on their worth to other consumers” (Zuckerman, 1999, p. 1406).

2.1 | Different routes to achieving product-category fit

Categorization reflects the acceptance of a new product's category membership by the category's customers (e.g., David & Lee, 2022). This acceptance can be difficult to secure when innovators and the customers hold differing perceptions of the innovation—a mismatch that may lead customers to reject the innovator's claims. Incremental innovations are often readily accepted by customers within an established category (e.g., Gregan-Paxton & Moreau, 2003), but more radical innovations may provoke disagreement about whether the product fits the category.

The existing literature identifies two distinct approaches for achieving product-category fit: creating a new category or fitting the innovation into an existing one.¹ The first approach may

¹Innovators can opt out of achieving product-category fit at the cost of delayed or denied acceptance of the innovation (e.g., Hsu, 2006; Hsu et al., 2009).

be especially appealing to innovators whose products diverge significantly from existing categories. In the absence of a suitable category, innovators may construct a new category to accommodate their product. This strategy has been explored in a variety of contexts, including architecture (Jones et al., 2012), haute cuisine (e.g., Durand et al., 2007), Indian art (e.g., Khaire & Wadhvani, 2010), microcomputers (e.g., Kennedy, 2008), nanotechnology (e.g., Granqvist et al., 2013), plant-based meats (e.g., Rhee et al., 2025), and satellite radio (e.g., Navis & Glynn, 2010). However, creating a new category can be costly and offers no guarantee of success (e.g., Rhee et al., 2017).

The second approach—fitting an innovation into an existing category—has received comparatively less attention. Scholars have proposed two strategies under this pathway: repositioning and redesign. Repositioning involves innovators responding to customer rejection by seeking category membership elsewhere, targeting a different category whose customers might find the innovation more relevant. Although research has considered the limits of firms' ability to reposition (e.g., Kovács et al., 2024), the practice is widespread in industries like pharmaceuticals. A well-known example is Viagra, a drug initially developed to treat heart problems, which was later repositioned to treat erectile dysfunction (Jourdan et al., 2020).²

Redesign, by contrast, involves changing the product itself to better align with existing category criteria. Zuckerman (1999, 2000), for example, studied how leaders of conglomerates—a relatively recent organizational form—attempted to resolve the firms' misalignment with financial analysts' industry categories. Because analysts often struggled to categorize conglomerates owning diverse, unrelated businesses, these firms received less coverage and traded at lower valuations than their more focused counterparts. To address this, some conglomerates redesigned their organizational structure, divesting unrelated divisions to enhance categorical clarity. However, we know little about what options remain for innovators when redesign fails to secure acceptance.

In summary, scholars have outlined two approaches through which firms commercializing innovations can achieve product-category fit: (1) creating a new category for the new product or (2) fitting the new product into an existing one. While the first approach has been studied across a range of industries, the second remains underexplored. Within this second approach, repositioning and redesign have emerged as hypothesized strategies. Yet, our understanding of how innovators fit their products into existing categories—and the broader implications of such efforts for category evolution—remains limited.

3 | RESEARCH SETTING AND METHODS

An ideal setting for developing such insight would allow us to observe how the process of fitting an innovation into an existing category unfolds over time. This process would start with an initial lack of fit between the innovation and the target category. From there, we could then observe customers' rejection of the innovation, the innovator's persistence in response to this rejection, and, ultimately, the innovation's eventual acceptance. Ideally, the context would also

²In the pharmaceutical industry, interest in repositioning—finding new indications for existing drugs—is driven partly by the high costs associated with discovering and developing new compounds, particularly the expense of clinical trials. Scholars in marketing and strategy have also considered product repositioning in response to changing consumer preferences (e.g., Anderson & Shugan, 1991; Villas-Boas, 2018) and evolving competitive environments (e.g., Seamans & Zhu, 2017; Wang & Shaver, 2014).



enable us to track customer evaluation criteria, the changes in the innovation's design, and how customers responded to those design changes.

The emergence of the market for MBS in the United States during the 1970s provides such a context. Major bond investors—especially pension funds, the innovation's target customers—rejected MBS for 13 years. During this period, MBS issuers introduced multiple generations of product redesign while also working to make MBS more comparable to existing bond products. Numerous archival sources document both the efforts of MBS issuers to gain acceptance among bond investors and the investors' responses. Moreover, the well-documented characteristics of financial innovations make it possible to precisely reconstruct product designs over time, including how these securities compared to bonds and how market participants evaluated them.

To examine how MBS issuers fit their products into the bond category, I conducted an historical case study. This method is particularly well-suited for investigating “organizational and social processes” and the “construction of organizations and markets” (Wadhvani & Decker, 2017, p. 122; Hargadon & Douglas, 2001), as well as for tracing the complex causal patterns in the innovation process (e.g., Cattani et al., 2017; Graham & Shuldiner, 2001).

To trace the evolution of the MBS market, I constructed an analytically structured narrative (e.g., Ingram, Silverman & Rao, 2012; Lubinski & Wadhvani, 2020; Rowlinson et al., 2014) using realist ontology (e.g., Vaara & Lamber, 2016). This approach involves reconstructing historical events using archival evidence that is proximate in time and context to events in question. It entails interpreting evidence from the perspective of historical actors, organizing events into coherent periods, and building a narrative that explains the observed trajectory. The method relies on abductive reasoning—comparing empirical evidence with conceptual and theoretical frameworks to derive the most plausible explanation. Rather than using step-by-step procedures to derive objective constructs, the methods of historical scholarship focus on systematically addressing the challenges posed by the structure and availability of the evidence (e.g., Wadhvani & Decker, 2017).

3.1 | Collecting evidence: Mapping the market structure and assembling the archival sources

The MBS market that emerged in the United States during the 1970s was shaped by reforms initiated in response to the Great Depression. In the 1930s, the decline in private-sector interest in financing mortgage loans prompted Congress to pass the National Housing Act, which established several federal agencies aimed at stimulating the flow of funds into the residential mortgage market. By the 1960s, the federal government sought to return the funding of residential mortgage lending to the private sector by encouraging institutional bond investors—particularly pension funds—to invest in mortgages (e.g., Ennis, 1960). However, investors resisted this shift. One reason was that the structure of residential mortgages had, over the decades, become more consumer-friendly and less aligned with investor preferences.³ The

³In the 1920s, a typical US residential mortgage had a 3–5-year term, during which borrowers made interest-only payments, followed by a lump-sum principal payment at maturity. By contrast, mortgages issued after the 1930s featured 20- to 30-year terms with monthly, level payments of both principal and interest (Fishback et al., 2013). This new structure, which gave consumers the option to repay their mortgages at any time, combined with the mortgage's long-term fixed interest rate made mortgages unattractive for bond investors (Jones & Grebler, 1961).

introduction of MBS was, in part, a response to this investor rejection of mortgages as suitable investments.

The 1970s MBS market comprised a complex ecosystem of participants with distinct roles. These included MBS issuers (e.g., mortgage lenders), bond investors (e.g., pension funds), intermediaries (e.g., investment banks), government agencies (e.g., regulators),⁴ and market observers. Table 1 describes the key groups of market participants.⁵

To capture the perspectives of these different actors, I collected a wide range of primary and secondary archival materials. I searched online databases including ABI Inform/Global, ProQuest Historical Newspapers, and WorldCat for documents related to organizations, individuals, and securities associated with the MBS market. These sources included security prospectuses, issuer and regulator annual reports, regulatory security listings, trade manuals, and promotional pamphlets. I also collected contemporaneous discussions of the challenges faced by market participants as documented in trade publications, popular press articles, and academic research. Furthermore, I contacted individuals active in the market during this period to obtain privately held documents not available in the public domain.

In total, my search yielded 741 publications, including 15 books and 726 industry documents and periodicals spanning the years 1960 to 2008. These sources enabled me to trace not only the evolution of MBS product design, but also changes in how bond investors evaluated these innovations. My analysis focuses on the period from 1970 (when MBS were first issued) through 1995 (when the process of redefining the boundaries of the bond category boundaries was effectively completed). Table 2 contains a representative list of the archival sources consulted.

3.2 | Interpreting the evidence

To interpret historical evidence from the perspectives of historical actors, historians rely on hermeneutic approaches. These entail: (1) preconception—the researcher's awareness of their own assumptions upon entering the archives, (2) distanciation—recognition of the temporal, cultural, and contextual distance between the researcher and the text, and (3) appropriation—the process of integrating historical insights into broader scholarly conversations (Wadhwani, 2023).

⁴(Quasi-)government agencies played multiple roles in the MBS market. Some agencies, such as FHLMC or FNMA issued MBS, while others provided guarantees—GNMA guaranteed the repayment of the securities themselves, and agencies like the Federal Housing Administration (FHA) and Veterans Affairs (VA) guaranteed the repayment of the underlying mortgages in the collateral pool. Additional agencies regulated the issuance of securities (e.g., Securities and Exchange Commission [SEC]), determining the tax implications for investors (e.g., Internal Revenue Service [IRS]), as well as setting relevant accounting standards (e.g., Federal Accounting Standards Board [FASB]). Although certain government agencies focused on regulating subsets of market actors—for example, the Department of Labor regulated pension fund investments, and the Federal Reserve oversaw some participating banks—these regulators played a less active role in the evolution of MBS.

⁵The different groups of actors collaborated while pursuing distinct but overlapping interests. Mortgage lenders issued MBS to secure a stable source of financing for underwriting mortgages. Investment bankers and other intermediaries promoted the acceptance of MBS because wider market acceptance translated directly into increased revenues. Government officials sought to (re-)privatize residential mortgage financing partly because of the challenges in supplying sufficient capital to finance residential mortgage lending, and partly because of a broader commitment to free market principles. Academics studied the mortgage market both to enhance its efficiency and to contribute to the advancement of knowledge through research and publication.



TABLE 1 Different types of MBS market participants and observers.

Participant type	Description
MBS issuers	Mortgage banks—mortgage lenders who originated the loans and sold them to long-term investors. Thrifts (e.g., savings and loan associations and mutual savings banks)—lenders who originated mortgage loans and held them in their own portfolios. Quasi-government agencies: Federal Home Loan Mortgage Corporation (FHLMC or Freddie Mac) and Federal National Mortgage Association (FNMA or Fannie Mae)—private companies chartered by US Congress to stimulate the development of secondary markets for conventional and government-insured mortgages, respectively.
Bond investors	Pension funds, insurance companies, commercial banks.
Intermediaries	Investment banks—Salomon Brothers, First Boston, and other banks structured, issued, and sold MBS consulting with mortgage lenders, government agencies, and investors. Trade manual publishers (e.g., Financial Publishing Company) published manuals used by the bond investors to compare securities.
Government agencies and regulators	Federal Housing Administration (FHA)—insured the repayment of mortgages Financial Accounting Standards Board (FASB)—set accounting standards. Government National Mortgage Association (GNMA or Ginnie Mae)—guaranteed the repayment of pass-through MBS backed by FHA- or VA-backed mortgages. Securities and Exchange Commission (SEC)—regulated the trading of publicly listed securities. US Treasury Department and Internal Revenue Service—set standards for tax treatment of securities. Veterans Administration (VA)—guaranteed the repayment of mortgages obtained by veterans.
Observers	Academics—economics, finance, and real estate faculty and graduate students. Press—freelance and staff writers for generalist and specialist publications.

I began collecting archival materials for this project in the aftermath of the 2008 financial crisis. Leveraging my prior professional experience in credit card and mortgage lending, I was initially interested in understanding how the acceptance of MBS as bonds might relate to the antecedents of the 2008 crisis. Amid a wave of foreclosures in 2007–2009, largely driven by borrowers' inability to repay loans, I was struck by a surprising historical contrast: during the early years of the MBS market, the central concern was not borrower default, but early repayment.

Rather than being a peripheral issue, the possibility of accelerated payments was a significant obstacle to investor acceptance of MBS as bonds. As Louis Nevins, director-counsel of the National Association of Mutual Savings Banks, noted: “The modified pass-through security is more like a bond than a mortgage, but the holder still has no protection against accelerated payments. Some potential purchasers are turned away by this feature” (Nevins, 1972, p. 42). In other words, because MBS could be prepaid ahead of schedule, they failed to meet a fundamental criterion of the bond category: predictable, long-term cash flows. Worries about prepayments

TABLE 2 Archival sources consulted.

Sources	Description
Trade press targeted at different market participants:	Mortgage lenders—<i>American Banker</i>, <i>Bottomline</i>, <i>Mortgage Banker</i>, <i>Mortgage Banking</i>, <i>Savings Bank Journal</i>, <i>Savings and Loan News</i>, <i>Real Estate Review</i> Bond investors—<i>Bond Buyer</i>, <i>Fixed Income Investor</i>, <i>Institutional Investor</i>, <i>Pensions & Investments</i>, <i>Pension World</i> Bank regulators—<i>Federal Reserve Bank of New York Quarterly Review</i>, <i>Review</i>—<i>Federal Reserve Bank of St. Louis</i>, <i>Federal Home Loan Bank Board Journal</i>
Academic research	Articles in <i>American Real Estate and Urban Economics Association (AUREA) Journal</i>, <i>Journal of Accountancy</i>, <i>Journal of Finance</i>, <i>Journal of Real Estate Finance and Economics</i>, <i>Real Estate Review</i>, <i>Regional Science and Urban Economics</i> Working papers—for example, NBER, Salomon Brothers Center for Study of Financial Institutions Doctoral dissertations and master's theses
Popular press	Major newspapers—<i>Wall Street Journal</i>, <i>New York Times</i>, <i>Washington Post</i>, <i>Los Angeles Times</i>, and <i>Chicago Tribune</i> General interest business magazines—<i>Business Week</i>, <i>Forbes</i>, <i>Fortune</i> Magazines focusing on investing—<i>Barron's</i>, <i>Crain's</i>, <i>Kiplinger's</i>, <i>Money</i>
Industry trade manuals and reference publications	All editions of the <i>Handbook of Mortgage-Backed Securities</i> (1985–2007) and <i>Handbook of Nonagency Mortgage-Backed Securities</i> (1997–1999) <i>Handbook of Securities of the United States Government and Federal Agencies</i> (1974–1990)
Legal statutes and rulings	Regulator reports—<i>Federal Register</i>, <i>Internal Revenue Cumulative Bulletin</i>, <i>HUD Statistical Yearbook</i> Congressional committee minutes and speeches

in MBS continued to shape the market for years to come (e.g., MacKenzie, 2011; Vinokurova, 2018a).⁶

This insight led me to investigate more deeply the evaluation criteria used by bond investors and how these evolved over time. While the archival record revealed persistent investor concern over prepayment risk, what I found particularly striking was that the degree to which MBS products protected investors against prepayments virtually unchanged throughout this period. This presented an empirical puzzle: if the securities' exposure to prepayment risk remained constant, why did bond investors ultimately come to accept MBS as bonds?

This puzzle suggested that product redesign alone could not account for the shift in investor acceptance of MBS. Instead, it raised the possibility that alternate mechanisms played a critical

⁶In engaging with the existing literature, I considered insights not only from management and historical scholarship but also from adjacent fields. The acceptance of mortgage securitization by financial market participants is well documented in the economic sociology literature (e.g., Carruthers & Stinchcombe, 1999; Davis & Kim, 2015; Krippner, 2012). However, through extensive reading, I realized that what remains less well understood is the process by which this acceptance was achieved.



role in enabling category fit. This finding shaped a central question of this study: how do innovators achieve product-category fit in the absence of substantive design changes?

3.3 | Reading against the grain

An enduring challenge in historical research is how to interpret the silence of the archives—that is, the systematic absence of certain voices or perspectives in the documentary record. Historians often address this problem by reading sources against the grain: identifying what is missing, interrogating why those omissions exist, and using available materials to reconstruct excluded viewpoints (e.g., Decker, 2013). In this study, the archival record predominantly reflects the perspectives of the innovation's proponents—a pattern consistent with the well-documented observation that financial innovations are typically sold rather than bought (e.g., Carroll & Lappen, 1994; Perry, 1979).

Between 1970 and 1995, pension fund managers—the target customers for MBS—were largely silent in the archival materials.⁷ This absence was not simply incidental but structural. A 1975 Harvard-MIT study found that many pension fund managers “may not be aware of the wide variety of mortgage-backed instruments now available” (Rosen, The mortgage market—High time America's pension funds moved in, *Pension World*, 1975, p. 22). This limited awareness was partly a function of institutional design: most pension funds outsourced their investment decisions to the trust departments of a small group of commercial banks, whose advisors themselves showed little interest in learning about MBS (Kirk, 1974; *Pensions & Investments*, 1975, Fund managers shun mortgages out of habit).

To reconstruct the perspectives of pension fund managers and how they evaluated MBS, I turned to indirect sources—specifically, the accounts of MBS issuers who struggled to sell these securities. These accounts provided insight into the evaluation criteria used by bond investors, as well as the solutions issuers developed in response to investor objections. From these accounts, I was able to infer the reactions of bond investors and the evaluation criteria they applied to MBS. One issuer described the conservatism of bond investors as follows: “The money boys just don't jump at new deals. They all like something that's been done before” (*Savings & Loan News*, 1977, Home mortgage-backed securities raise new capital for lenders, p. 22). Similarly, the GNMA MBS Dealers Association cited the “antipathy of the investment community to a new security with characteristics that deviate from the norm” (GNMA Mortgage-Backed Securities Dealers Association, *The Ginnie Mae Manual*, 1978, p. 21). Throughout this analysis, I tracked the provenance of each quotation and perspective by citing both the publication and organizational affiliation where available.

⁷My broader searches of the documents in the ABI ProQuest databases yielded only a handful of articles containing direct quotes from pension fund managers. To supplement these results, I searched the archives of *Pensions & Investments* (*Pensions & Investment Age* 1981–1989), *Pension World* (*Pension & Welfare News* pre-1975), looking for articles published between 1970 and 1996 that contained the word “mortgage.” This search yielded 86 articles (compared to 56 yielded by the broader search). Of these, 46 referenced MBS; the remainder focused on commercial mortgages or real estate more broadly. Among the 46, one article was co-authored by a pension fund manager, one included a quote from a pension fund manager explaining the rationale for purchasing a particular type of MBS, and one featured a full-length interview with a pension fund manager. The remaining 43 publications were authored by lawyers, investment bankers, or mortgage bankers seeking to convince pension fund managers to invest in the mortgage market.

To further illuminate how investors defined the bond category, I examined contemporaneous industry advice on evaluating bonds. A particularly influential source was Sidney Homer, a managing partner at Salomon Brothers and a leading authority on bond investing. In a 1962 *Barron's* article titled “How to buy bonds” Homer outlined key evaluation criteria: quality (likelihood of default), maturity (repayment schedule), yield (expected return), call protection (restrictions on early repayment), and marketability (ease of resale) (1962, p. 5).

Of these, the marketability criterion is especially relevant for understanding investors' resistance to MBS. Specifically, it reflects the social nature of financial products: the value of bonds “depends heavily on their worth to other consumers” (Zuckerman, 1999, p. 1406). Categorization, therefore, plays a crucial role not just in individual assessment, but in shaping the collective confidence needed to support the market. Viewed through this lens, efforts to position MBS as bonds were not only about aligning product features, but also about signaling broad category membership to assure investors of future demand.

3.4 | Periodization and explanation

One challenge in interpreting MBS proponents' descriptions of market developments is their tendency to claim success at every step of the way. From the first issuance of MBS onwards, mortgage lenders and the investment bankers made a point of involving at least one pension fund in the initial distribution round and then proclaiming broader success in winning over pension funds. Any increase in pension fund interest was interpreted as a definitive breakthrough. Some of this triumphalism was strategic, occasionally bordering on euphemistic. For example, Bank of America's first private MBS issue in 1977, which saw limited demand, was described as having been “nearly sold out within hours” (*Pensions & Investments*, Bright future seen for mortgage device, 1977, p. 10). Other overstatements likely stemmed from a lack of historical perspective; every increase in the market size appeared, at the time, to be the most significant yet.

To address this tendency of overclaiming success, I compared the claims found in archival documents with broader market performance data, including issuance volume and the proportion of MBS bought by pension funds. Another telling indicator of genuine progress was the cessation of victory claims—once pension funds began to consistently accept MBS as bonds, the issuers and (quasi-)government agencies no longer felt the need to advertise pension fund participation. For instance, the Federal Home Loan Mortgage Corporation (FHLMC or Freddie Mac) and the Government National Mortgage Association (GNMA or Ginnie Mae) stopped reporting pension fund MBS holdings in their annual reports in 1983 and 1987, respectively, suggesting that pension fund participation had become normalized.

I used the introduction (and in some cases, failure) of different MBS designs to separate the history of the market into three distinct periods: (1) 1970–1973: Design and redesign, which included the launch of the first MBS in 1970, a major redesign in 1972, and the final issue of the redesigned MBS in 1973; (2) 1974–1982: Commensuration efforts, a phase marked by attempts to make MBS more comparable to bonds; and (3) 1983–1995: Acceptance, when MBS—starting with Collateralized Mortgage Obligations (CMOs)—began to be widely accepted as bonds by pension funds. While pinpointing the exact moment of acceptance is difficult, evidence of early acceptance can be seen in pension funds buying the majority of CMOs starting with the securities' introduction in 1983. Markers of this turning point include the disappearance of pension fund MBS holding statistics from the issuers' annual reports in the mid-1980s



and, by the 1990s, changes in how financial trade press dictionaries and regulators categorized MBS.

In keeping with this periodization, my analysis centers on three types of MBS: pass-through certificates (the original MBS design), serial notes (the first redesign aimed at matching bond investors' expectations), and CMOs (the first MBS widely accepted as bonds).⁸ In analyzing these securities, I tracked their design characteristics, how they compared to prior instruments, and how bond investors evaluated them. In developing this historical case study, I iterated between the narrative emerging from the case and the archival evidence to abductively derive the main factors driving MBS evolution and eventual acceptance.

My findings describe the process by which market participants succeeded in fitting MBS into the bond category. I start by explaining the rationale behind MBS and the reasons for bond investor rejection. I then discuss the efforts to make MBS more comparable to bonds, focusing on three strategies: (1) Redesigning MBS to more closely resemble existing bond products; (2) Finessing yield and maturity estimates to satisfy bond investors' evaluation criteria; and (3) Altering the boundaries of the bond category itself to accommodate MBS. I conclude by considering alternative explanations and presenting two sources of evidence pointing to a change in category boundaries: changes in customer behavior and shifts in market participants' beliefs.

4 | FITTING MBS INTO THE BOND CATEGORY

4.1 | Historical context: Rejection of mortgages by pension funds

The MBS market that emerged in the United States in the 1970s was shaped in part by the federal government's efforts to return the funding of residential mortgage lending back to the private sector. Specifically, the government sought to encourage bond investors (i.e., pension funds) to invest in mortgages (e.g., Ennis, 1960). However, these efforts initially failed because investors viewed mortgages as inferior to bonds—citing, for example, the significantly higher staffing costs associated with managing mortgage investments (e.g., Kanner, 1982).⁹

In response to pension funds' lack of interest in mortgage investing, the 1968 Housing and Urban Development Act laid the groundwork for the creation of new securities backed by mortgages and designed to attract bond investors' capital to the mortgage market. The Act authorized the Government National Mortgage Association (GNMA) to guarantee the repayment of securities collateralized by federally insured mortgages.¹⁰ The first generation of these securities was the GNMA pass-through certificates (so called because they “passed through” the mortgage payments from borrowers directly to investors). Figure 1 offers a schematic representation of the GNMA pass-through securities.

First issued in 1970, pass-through certificates were designed to simplify mortgage investing. The certificates entitled investors to a share of cash flows from a pool of mortgages that served

⁸For a detailed treatment of other MBS designs in this market see Vinokurova (2018b, 2019). For an analysis of how MBS designs in the 1970s market in the United States compare to those of the 1870s and the 1920s markets, see Vinokurova (2020).

⁹This was due to individual mortgage loans (1) being smaller in denomination compared to bonds, (2) featuring idiosyncratic documentation, and (3) requiring the investors to collect payments from mortgage borrowers.

¹⁰The Act created GNMA as the part of the Federal National Mortgage Association (FNMA or Fannie Mae) that would remain a federal agency after the rest of FNMA was spun off as a quasi-private corporation.

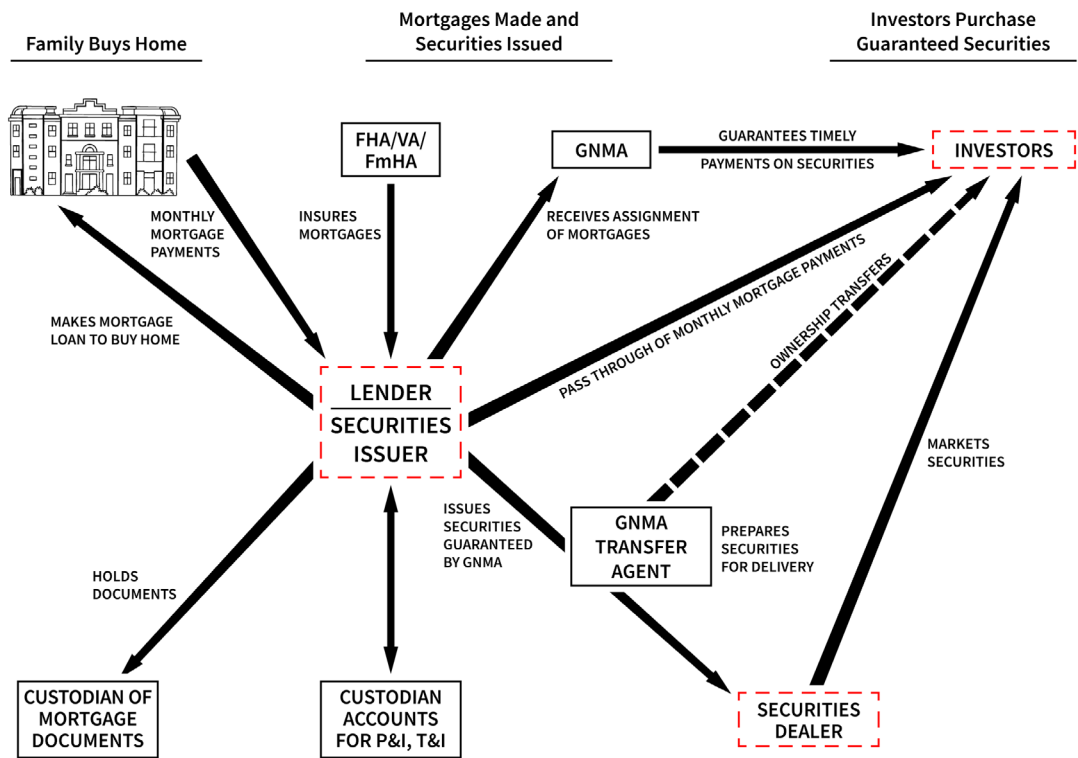


FIGURE 1 Structure of GNMA mortgage-backed securities. *Source:* Government National Mortgage Association, *Annual Report*, 1981, p. 5.

as the securities' collateral. Early MBS investors argued that pass-through certificates reduced the staffing costs of investing in mortgages by a factor of 50 (e.g., Adams et al., 1971).¹¹

4.2 | Early MBS design: Positioning MBS as bond substitutes

Mortgage lenders marketed pass-through certificates as substitutes for bonds. In the words of Bundy Colwell—chairman and president of Colwell Co., “the nation’s fifth-largest mortgage banking firm, the most active issuer of mortgage-backed certificates”:

They reduce costs and provide a lot more flexibility... More and more, this is the way mortgages will be marketed. They will tap markets, particularly pension funds and other institutions, that are not equipped to handle mortgages as such. *All they have to do is treat them like bonds.* [emphasis added] (Loehwing, 1970, Fat years and lien, *Barron's*, p. 3).

¹¹The certificates addressed all three major objections bond investors had to mortgage investing: (1) the small denomination, (2) the idiosyncratic documentation of individual loans, and (3) the need for the investors to collect payments from individual borrowers, thus reducing the staffing costs associated with mortgage investing.



The positioning of MBS as bond substitutes reflected the belief that doing so would help attract bond investors' capital. James Draper of the investment bank Pressprich Inc. put it succinctly: "Much more money, some will argue, could be made available to the mortgage market if pension fund managers would think of Ginnie Maes as simply another type of bond" (Draper, Ginnie Maes—More attractive than ever, *Pension World*, 1976, p. 29). The creation of these securities was thus guided by mortgage lenders' interest in attracting bond investors—especially pension funds—to the mortgage market. This motivation was captured visually on the cover of *Mortgage Banker*, shown in Figure 2:

However, for bond investors to treat MBS as bonds, MBS issuers had to address an important difference between the two: the lack of predictability in the timing and amount of pass-through cash flows. While bondholders typically received fixed semiannual interest payments and a single principal payment at maturity, MBS investors received monthly payments of principal and interest, the amounts and timing of which vary until the loan is paid in full. Because mortgage borrowers can repay any portion of their principal at any time without notice, cash flows from pass-through certificates are inherently uncertain.

This unpredictability posed two challenges to marketing pass-through certificates as bonds: (1) it made it difficult to identify bonds of comparable maturity, and (2) it complicated the calculation of expected investment returns. To address the first issue, MBS issuers promoted the assumption that the underlying mortgages would, on average, be repaid within 12 years. This allowed the trade press to describe pass-through certificates as "what amounts to a 12-year government bond" (Loehwing, 1970, p. 3).

To resolve the second issue—uncertain returns—issuers introduced a new metric to supplement the traditional bond measure of yield that they described as "yield to full maturity" with "yield to estimated maturity," security return under the 12-year assumption. Walter Kingman, then president of GNMA, illustrated this approach:

That meant a yield to the investor on the full [stated] 25-year maturity—and the maturity is based on the longest length of mortgage in the pool—of 8.58 percent.¹² On a more realistic 12-year basis—as you know mortgages are prepaid and a 12-year maturity is a realistic, historical pattern—the investment yield was 8.75 percent. In today's market, that's quite good (Mortgage-backed security, *Mortgage Banker*, 1970, p. 67).

As this example shows, using a 12-year estimated maturity allowed issuers to present MBS as offering superior returns compared to their performance on the traditional bond yield measure.¹³

In summary, MBS were specifically designed to attract bond investors' capital into the mortgage market by being marketed as bond substitutes. To make this comparison viable, issuers introduced estimates of MBS maturity and yield metrics to compensate for the unpredictability of MBS cash flows.

¹²A pass-through certificate would remain outstanding until the last loans were paid in full.

¹³The two yield measures—yield to 30-year (i.e., stated) maturity and yield to 12-year (i.e., estimated) maturity—became the commonly accepted yield measures for pass-through certificates (White, 1976, p. 11; GNMA Mortgage-Backed Securities Dealers' Association, 1978, pp. 11, 12).



FIGURE 2 Fitting mortgages into pension fund portfolios.



4.3 | Rejection, redesign, and rejection: Bond investors do not buy MBS

The design of pass-through certificates significantly reduced the cost of investing in residential mortgages compared to buying individual mortgage loans. Despite these cost savings—and the aggressive marketing of MBS as bond substitutes—pension funds showed little interest in pass-through certificates.

A key reason was the unpredictability of the repayment patterns of the underlying mortgages. Observers frequently emphasized this issue, describing the pass-through certificates as having “irregular” principal payments (*Wall Street Journal*, Government weighs S&L bond sale idea as a way to lessen the mortgage pinch, 1973, p. 2), being “fraught with monthly accounting problems” (Ganis, All about GNMA mortgage-backed securities market, *Real Estate Review*, 1974, p. 61), and featuring “unpredictable amounts of principal and interest” (Sojacy, Mortgage securities outshine others, *Pensions & Investments*, 1980, p. 24). One commentator went so far as to label them an “accounting nightmare” (Quint, Your money: Pass-through securities, *New York Times*, 1981, p. 34). These concerns about lack of predictability in the MBS cash flow patterns emerged as a central obstacle to bond investors’ acceptance of the securities.

Specifically, MBS issuers struggled to attract investors with long-term horizons, such as pension funds, because the investors feared that their holdings might be repaid in just 12 years (or less) rather than the 25 or 30 years suggested by the securities’ stated maturities (e.g., Kidd, 1970). Moreover, investors with shorter-term horizons were unwilling to invest in securities with such long stated maturities (e.g., Fink, 1996; Snoddy & Dickinson, 1971). In short, the uncertainty surrounding repayment patterns effectively left MBS with no customers: the securities’ maturities were too short for long-term investors, too long for short-term investors. This uncertainty prevented bond investors from accepting MBS as bond substitutes.

4.3.1 | Redesign to better fit the existing category: MBS with narrower ranges of maturities

By 1971, MBS issuers had come to terms with bond investors’ lack of interest in pass-through certificates and began proposing redesigned securities to address the objections (Jones, 1971). The first such redesign was the GNMA Serial Notes, issued in 1972, which aimed to make MBS more “bond-like” by offering investors a narrower and more predictable range of maturities.

Unlike pass-through certificates, which distributed borrower payments to all investors simultaneously, serial notes repaid investors sequentially; each certificate had a different, more narrowly defined, maturity. The Financial Publishing Company, a publisher of yield tables used by bond investors to compare securities, described the new design in an advertisement:

GNMA Serials... A new type of mortgage-backed security has now been dreamed up which we call “Serial” type by analogy with a serial bond issue. In most [pass-through certificate collateral] pools all the bond holders own pro-rata all the mortgages. On \$5,000,000 pool of 30-year 7s, represented by 200 certificates of \$25,000, the principal repaid the first year (\$51,300) gives \$256.50 to each certificate holder. Not so on the serial type. Certificate #1 gets all the principal first, then #2, etc. The holder of certificate #25, for example, knows that he must be paid off in 10 years (not 30 years). FHA prepayment experience would actually indicate a 4-year likely

term for him (Financial Publishing Company, GNMA Serials, *Mortgage Banker*, 1972a, p. 30).

This design effectively divided the 30-year stated maturity of the mortgage pool into 200 periods, the length of which depended on borrower payments. The goal was to accommodate both long-term investors, who were wary of early repayments, and short-term investors, who were deterred by long stated maturities. Louis Nevins, director-counsel of the National Association of Mutual Savings Banks, articulated the promise of serial notes:

The purpose of the serial note security is to attract with a single issue investors with varied maturity requirements. Hence an investor interested in a five-year security could be accommodated in the same issue as an investor interested in a much longer term issue (Nevins, Mortgage-backed securities—A multifaceted investment tool, *Savings Bank Journal*, 1972, pp. 28, 32).

Nevins also expressed the hope that by offering some protection against accelerated payments, serial notes would narrow the trading spread between MBS and bonds—attesting to greater substitutability between the two.

4.3.2 | Rejection of the redesign

Despite these intentions, GNMA Serial Notes failed to gain traction among bond investors. They were discontinued after only two years, with just \$130 million issued—less than 2% of the GNMA pass-through certificates issued over the same period. A *Pension & Welfare News* article jointly authored by a pension fund manager and a Salomon Brothers director of investment services attributed the persistent difficulty in attracting bond investors to the absence of “a consistent common framework” for comparing bonds and mortgages and, by extension, MBS (Leibowitz & Lechner, Measuring and managing the mortgage vs. bonds yield relationship, 1974, p. 22). This diagnosis pointed to a broader challenge: MBS were not only different from bonds but also existed outside the conceptual boundaries of the bond category. Addressing this problem would require efforts not just to redesign MBS, but to reshape the evaluative framework used by investors—ultimately changing the boundaries of the bond category itself.

4.4 | Making MBS more comparable to bonds

4.4.1 | Transforming prepayments into risk

Faced with the rejection of both pass-through certificates (the original MBS design) and serial notes (the first attempt at MBS redesign), market participants focused on equipping investors with more accurate tools for estimating MBS maturity and yield. The goal was to make MBS more comparable to bonds. Lewis Ranieri, a pioneer in the MBS trading space, captured the MBS traders' early frustrations: “We were not able to convey to customers the average life of securities because none of us really knew how to compute it accurately” (Origins of securitization, *A primer on securitization*, 1996, p. 36).



At the heart of the problem was mortgage prepayment behavior, which made it difficult to predict when investors would be repaid. Unlike bonds, whose repayment schedules were fixed, MBS were exposed to the whims of individual borrowers who could repay their loans at any time. Accurately estimating prepayments required fine-grained data on how borrowers responded to changes in prevailing interest rates. Over the course of the 1970s, MBS issuers (e.g., Struck, 1974), investment banks (e.g., Connolly, 1977), and finance academics (e.g., Peters, 1979) collected data on this relationship. By 1978, the once-standard assumption of a 12-year maturity estimate was supplemented by a range of scenario-based maturity estimates. These scenarios allowed analysts to calculate corresponding yield ranges for each pool of mortgage collateral. For example, traders at First Boston introduced what they termed “honest-to-God yield” and “honest-to-God-corporate-bond-equivalent-yield” for individual pass-through certificates (Moffitt, Ginnie Mae pass-through certificates offer higher yields plus safety for cautious investors, *Wall Street Journal*, 1978). These calculations allowed for more precise comparisons with bonds, accounting for the variability introduced by prepayments. To further aid the comparability of MBS and bonds, Salomon Brothers introduced its *Total Rate of Return for Mortgage Pass Throughs Index* in 1979 to provide “definitive statistics on paydown experience across the various pass through sectors” and offer “a comparison to returns in the corporate bond market” (*Wall Street Journal*, Display Ad 103, 1979, p. 39).

This growing sophistication in modeling repayment behavior enabled the transformation of the prepayment uncertainty into risk. This transformation changed the view of prepayments from unknowable to quantifiable. The result was a new risk: prepayment risk, defined as—the risk “that prepayments may not occur at the rates that had been anticipated” (Senft, The “true yield” of a pass-through security, *Mortgage Banker*, 1979, p. 19). Unlike traditional bonds, MBS carried this additional dimension of risk—originating from the behavioral dynamics of early repayment. With this shift, the unpredictability that had previously made MBS incomparable to bonds was reframed as a manageable, measurable risk. This marked a critical step in making MBS intelligible within the evaluative frameworks of bond investors.

4.4.2 | Aligning prepayment risk with call risk—The risk of bonds being repaid before maturity

Prior to the introduction of MBS, investors treated mortgage prepayments and early repayment of bonds as unrelated phenomena. For instance, one of the earliest studies of investor attitudes toward early bond repayment found that “Lenders who have expressed great concern over the redemption features of corporate bonds have made residential mortgage loans with liberal prepayment privileges” (Hess & Winn, *The value of the call privilege*, 1962, p. vii). This disparate treatment stemmed from investors’ beliefs that bond issuers and mortgage borrowers had fundamentally different motivations for early repayment. Specifically, early bond repayment decisions were viewed as primarily driven by changes in interest rates, while mortgage prepayments were “believed to be relatively insensitive to interest rate effects” (Leibowitz & Lechner, 1974, p. 22).

However, the collection of mortgage prepayment data—prompted by the introduction of MBS and the efforts to market them to pension funds—challenged this dichotomous view. Academics and practitioners studying this data arrived at a more nuanced understanding of mortgage prepayments and their sensitivity to interest rates. In particular, they documented an overlap between bond issuers’ and mortgage borrowers’ reasons for early repayment. Mortgage

borrowers, they concluded, prepaid their mortgages both for interest-rate related reasons (i.e., drops in the interest rate levels) and for non-interest-rate-related reasons (such as job-related relocations or changes in family size).

This shift in understanding of prepayments was articulated by Richard Haney, a professor of real estate at the University of Georgia, in a 1978 article comparing yields on MBS and corporate bonds: “The *portion of the callability risk common to both mortgages and bonds varies with the level of interest rates*. The lower the market rates, the greater the probability of refinancing for older, higher-rate mortgages and bonds” [emphasis added] (An analysis of yield spreads between Ginnie Mae pass-throughs and Aaa corporate bonds, *Financial Management*, 1978, p. 21). Haney’s statement illustrates how prepayment risk and call risk—once treated as distinct—were now mapped on a common dimension: early repayment risk.

This dimension was divided into two components: call risk (early repayment for interest rate reasons) and prepayment risk (early repayment for non-interest rate reasons). By conceptualizing prepayment risk as a form of call risk, market participants were able to situate MBS more clearly in the bond category.

Figure 3 illustrates how aligning prepayment risk with call risk altered perceptions of similarity between MBS and bonds.

4.5 | Evidence of changes in category boundaries: Bond investors buy MBS

In 1983, the redefinition of the boundaries of the bond category culminated in pension funds buying a new type of MBS: collateralized mortgage obligations (CMOs). Freddie Mac, the quasi-government agency that issued the first CMOs, described the success of the securities in its 1983 annual report:

Of the \$1.7 billion in CMOs Freddie Mac issued in 1983, pension funds were the largest single direct investor, obtaining nearly 30 percent. Commercial bank trust departments, many of which manage pension assets, were the next largest investor at 18 percent (p. 6).

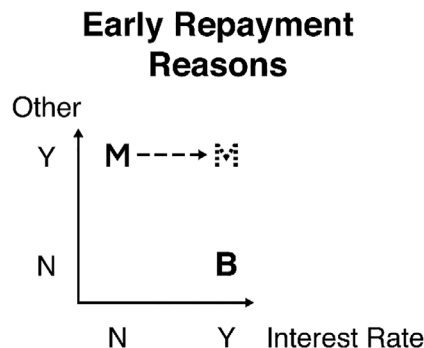


FIGURE 3 Aligning prepayment risk in MBS with call risk in bonds.



This 30% share exceeded pension funds' historic holdings of pass-through certificates, which never rose above 12% in any given year prior to the introduction of CMOs.¹⁴

Like GNMA Serial Notes—the 1972 attempt at MBS redesign—CMOs offered multiple securities with different maturities backed by a single pool of mortgage collateral. Kenneth Thygerson, then-president of Freddie Mac, explained the rationale behind the new securities: “The collateralized mortgage obligations are designed to reduce the uncertainty associated with the timing of principal payments, and thus to be attractive to a wider range of investors” (*New York Times*, Mortgage agency to sell collateralized securities, 1983, p. D7).¹⁵ Similarly, *A Barron's* reporter explained that CMOs were “structured to confront a major objection that fixed-income investors have leveled at most mortgage-backed securities—the absence of call protection” (Meagher, A Bond's Encore, 1983, p. 84).

By 1992, the success of this strategy was evident. A *Pensions & Investments* reporter described the MBS market as the second largest segment of the bond market, behind only US government bonds (Star, Use of mortgage-backed doubles, 1992). The *Wall Street Journal* attributed this transformation to a shift in investor beliefs:

[The CMOs] reduced the risk of prepayment for holders of the longer-maturity bonds by earmarking all early mortgage retirement payments for holders of the shorter-maturity bonds. Convinced by this device that the bonds were as reliable as many corporate issues, investors snapped them up (Conte & Schellhardt, Big secondary market in mortgages smooths flow of housing funds, 1983, p. 16).

Market observers emphasized that CMOs succeeded with pension funds specifically because they offered call protection:

The big coup for the new securities was the pension funds, whose \$600 billion in assets have long made them sought-after candidate for pumping more money into the housing industry. Because the funds must match pension liabilities that will not come due for several years against some long-term asset, the risk of getting paid off too early had caused them to steer clear of mortgage investments (*Business Week*, Making mortgages more appetizing for investors, 1983, pp. 170, 171).

Henry Kaufman, chief economist at Salomon Brothers, directly linked the CMOs' bond status to the increased predictability of their repayment timing: “In 1983, the securitization of the mortgage market was spurred by the introduction of mortgage-backed bonds that offer investors greater certainty of cash flow through the implementation of several maturity classes” (*Interest rates, the markets, and the new financial world*, 1986, p. 175).

The acceptance of CMOs by bond investors propelled explosive growth in the MBS market:

Experts estimate that \$1 trillion in new and existing mortgage securities were traded in 1985, 10 times the 1980 trading volume of slightly over \$100 billion. Major

¹⁴In its annual reports, Freddie Mac reported pension funds' holdings of pass-through certificates growing from 9.2% in 1977 to 11% in 1982. GNMA reported pension funds' holdings of pass-through certificates increasing from 5.39% in 1972 to 9.1% in 1982.

¹⁵Freddie Mac was a nickname Wall Street traders gave to Federal Home Loan Mortgage Corporation (FHLMC). They dubbed GNMA—Ginnie Mae and Federal National Mortgage Association (FNMA)—Fannie Mae.

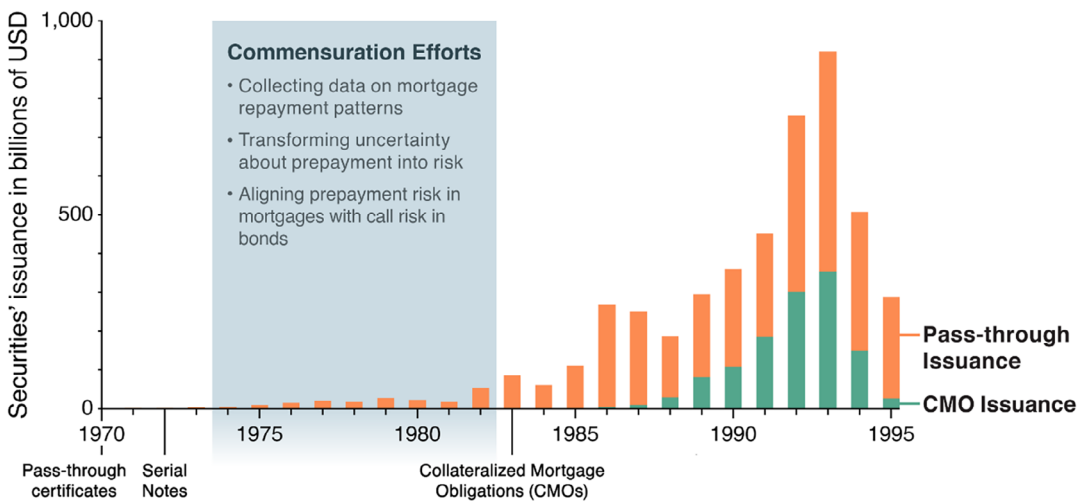


FIGURE 4 MBS issuance timeline. The growth in the CMO issuance contributed to the growth in the issuance of pass-through certificates, in part, directly as pass-throughs were used for CMO collateral; however, the increase in the pass-through issuance outpacing the issuance of the CMOs is consistent with the possibility that the acceptance of CMOs as bonds also promoted a broader acceptance of pass-throughs as bonds.

institutional investors—pension funds, investment advisers, the trust departments of commercial banks, and so forth have become major players in the mortgage securities market, which in 1985 equaled one-fifth of the total market in taxable, investment-grade bonds (Federal Home Loan Mortgage Corporation, *Annual Report*, 1985, pp. 21, 22).

Figure 4 shows the relative growth in CMOs and pass-through issuance between 1970 and 1995.¹⁶

The investors' contrasting responses to CMOs and GNMA Serial Notes—despite their design similarities—attest to the efficacy of MBS issuers' efforts to redefine the bond category between 1973 and 1983. Like serial notes, CMOs featured multiple securities structured for sequential repayment, each with its own yield and maturity estimates, and all backed by a single pool of mortgages. Both were marketed with emphases on the same advantages.

Table 3 compares the similarities in trade and popular press descriptions about GNMA Serial Notes and CMOs.

The different reception of these two securities was not due to the structure of the securities. Rather, this change stemmed from a shift in the conceptual framework through which they were evaluated. In short, the boundaries of the bond category were altered during the period between the two securities' introduction. By transforming mortgage prepayments into prepayment risk—and aligning this new risk with the familiar call risk—MBS market participants expanded the boundaries of the bond category. CMOs, marketed as MBS with built-in call protection, were now considered as legitimate fixed-income investments. This shift in perception is evident in their rapid adoption: serial notes' issuance amounted to just \$130 million over two

¹⁶ Attesting to pension funds' acceptance of MBS, Freddie Mac and GNMA stopped including pension fund MBS holding statistics in their annual reports in 1983 and 1987, respectively.



TABLE 3 Similarities in claims between serial notes and CMOs.

	GNMA serial notes	CMOs
Predictable final maturities for the short-term investor	There is a new breed of issue in the pass-thru area whereby the mortgage pool is sliced into uniform pieces, each with a fixed maximum maturity, just like a bond (Financial Publishing Company, <i>Passing Thru, Mortgage Banker</i> , 1972b, p. 13). A serialized pass-through security offers issuers an opportunity to sell a short-term instrument to short-term investors, thereby expanding the types of investors that can be attracted to this form of mortgage investment (Kidd, One year-old and going strong, <i>Mortgage Banker</i> , 1971, p. 40).	Holders of the first class of collateralized mortgage obligations thereby have the certainty of minimum sinking fund payments that offer a definite maximum maturity (Sloane, New version of Ginnie Mae, <i>New York Times</i> , 1983, p. 38). You can convey to the short-term buyer an absolutely short-term instrument (<i>Savings Institutions</i> , FHLMC Three-Tiered Bond Scores Big, 1983, p. 118).
Protection from prepayment	The serial note program protects the holder to some extent against an accelerated payment—particularly on single-family pools (Nevins, Mortgage-backed securities—A multifaceted investment tool, <i>Savings Bank Journal</i> , 1972, p. 42).	This type of offering provides more call protection than the traditional mortgage-backed security (Zigas, Corporate issuers face competition, <i>Bond Buyer</i> , 1983, p. 3).
Range of maturities	The Serial Note concept of a mortgage-backed security... was developed to offer investors a wide selection of maturities within a pool (Government National Mortgage Association, <i>Annual Report</i> , 1973, p. 7).	The obligations allow Freddie Mac to pool mortgages and offer investors different classes of maturities (Zigas, Corporate issuers face competition, <i>Bond Buyer</i> , 1983, p. 3).

years, while CMOs exceeded \$9.5 billion in issuance within nine months. Moreover, CMOs quickly became a substantial component of corporate bond indices, used to evaluate bond investors' performance (Hamecs, 1984).

4.5.1 | Alternative explanations to changes in category boundaries: Did CMOs eliminate prepayment risk?

While it may appear that CMOs were accepted as bonds primarily because of their improved design—specifically, more effective protection from early repayments and more predictable cash flows—contemporaneous evidence and subsequent investor behavior suggest that structural changes alone were insufficient to account for their success. Crucially, the limitations of CMO design with respect to managing prepayments were recognized shortly after their launch:

What it [the CMO] does not solve is the prepayment problem. Let's say you create a CMO with 15 percent mortgages and everybody refinances for savings and debt service. Every series gets retired immediately. Simply put, a CMO is not foolproof protection against prepayment if those prepayments are being driven by massive

refinancing for savings and debt service. If they come from simple, normal demographic mortgage turnover then that's fine because that's what is imputed into the maturities. In other words, the 12-year life assumption. But if you accelerate that process by massive financings, everybody gets blown out (*American Banker*, Interview: Lewis S. Ranieri, 1985, p. 22).

Ranieri's admission reveals that CMOs did not substantively improve upon the predictability of MBS cashflows beyond the 12-year life assumption introduced in 1970. By the 1990s, the same unpredictability in repayment timing that once excluded MBS from the bond category was now interpreted as proof of their category membership.

For instance, in response to the refinancing wave of the early 1990s, when CMOs failed to protect investors from prepayment, the setback did not trigger a reassessment of MBS. On the contrary, it confirmed their status as part of the bond category. A *Bank Management* reporter explained: "MBS is the ultimate callable bond. There's risk associated with them. You don't get spreads over Treasuries for free" (Borowsky, Banks struggle to control mortgage prepayment risk, 1993, p. 42). This reaction demonstrates that by the 1990s, investors no longer expected MBS to eliminate prepayment risk. Instead, they had come to accept prepayment risk as an extension of call risk. Thus, rather than undermining the inclusion of MBS in the bond category, the continued presence of prepayment risk actually affirmed it. In this light, the acceptance of CMOs as bonds is best understood not as a response to their innovative design, but as the culmination of the commensuration efforts that made MBS comparable to bonds.

4.6 | Evidence of changes in category boundaries: Changes in market participants' beliefs

If CMOs were not accepted as bonds due to their greater efficacy at managing prepayment risk, what explains their eventual acceptance? Beyond changes in bond investor behavior, archival sources provide evidence of a deeper shift: a transformation in how market participants conceptualized what it meant to be a bond.

Contemporaneous trade dictionaries document an evolution in the definition of the bond category. For instance, the 1996 edition of the *Dictionary of Financial Risk Management* defines a bond as follows:

Traditionally, a written unconditional promise to pay a specific principal sum at a determined future date, and interest at a fixed or determinable rate on fixed dates. Increasingly, the promise to pay has become conditional, and the principal, interest, and payment dates have become contingent in real world instruments (Gastineau & Kritzman, 1996, p. 37).

By contrast, the 1992 edition of the same dictionary—published just 4 years earlier to “codify the financial risk management lexicon” (Gastineau, 1992, inside cover)—did not contain an entry for “bond,” implicitly assuming that the category was self-evident and required no definition.

To appreciate the change in the dimensionality of market participants' representations of the bond category, consider how the risks inherent in bond investing were described before and after the acceptance of MBS into the bond category. In 1979, Dexter Senft, a vice president at



the First Boston Corporation, explicitly distinguished the risks of “ordinary bonds” from those of MBS:

Investors in ordinary bonds face *two* types of risk. The first is credit risk, which is the risk that an issuer may default at some future date. The second is market risk, which is the risk that prices will change. This is important whenever returns are measured for holding periods that are shorter than maturity. Pass-through [MBS] investors face a third kind of risk called prepayment risk. [emphasis added] (“True yield” of a pass-through security, *Mortgage Banker*, 1979, p. 19).

In this conceptualization of the bond category evaluation criteria, prepayment risk is explicitly excluded from bond category dimensions—it is something that sets MBS apart from bonds.

By 1995, however, this distinction collapsed. Peter Goldman, a managing director at Bear, Stearns & Co., described prepayment risk as one of the core risks inherent in bonds:

The potential damage done to bondholders by the call features of callable bonds. These are bonds that can be terminated prematurely at the option of the borrower. The group includes mortgage-backed securities like Ginnie Maes, many corporate bonds, and some Fannie Mae and Freddie Mac bonds (Sell callables, buy Junk, *Forbes*, 1995, p. 256).

In this revised conceptualization, prepayment risk is no longer a disqualifying feature—it is a risk inherent in bond investing. These rhetorical and conceptual shifts signal the inclusion of MBS and prepayment risk into the bond category.

The recognition of this redefinition was not limited to practitioners. Financial market regulators also acknowledged the prepayment risk dimension as part of the bond category. For instance, a 1991 accounting rule proposal by the Federal Accounting Standards Board (FASB) characterized both MBS and callable bonds as “investments with prepayment risk” (Journal of Accountancy, FASB issued ED on prepayment risk, 1991, p. 27). Similarly, a 1995 request for comments on a new policy by the Securities and Exchange Commission described MBS as “bonds with prepayment options” (Improving disclosure of risk by mutual funds and other investment companies, *Federal Register*, 1995, p. 17175). The interpretations of MBS in the 1990s by regulators who did not actively participate in promotion of MBS to bond investors reflect the fact that MBS issuers did not merely redesign or reposition their products—they redefined the bond category boundaries to accommodate MBS.

4.7 | Findings summary

This paper considers an extreme case of the challenges innovators face in fitting a new product into an existing market category. Far from being unproblematic, the process was marked by prolonged resistance to the claim that MBS were “like bonds.” When redesigns failed to resolve the mismatch between product features and category expectations, MBS issuers pursued a different strategy: they altered the meaning of the category. This study shows how innovators successfully modified the dimensions of the bond category by first transforming mortgage prepayments into a type of risk and then aligning the prepayment risk with an existing category dimension—call risk. This alignment allowed the investors to view MBS as a type of bonds,

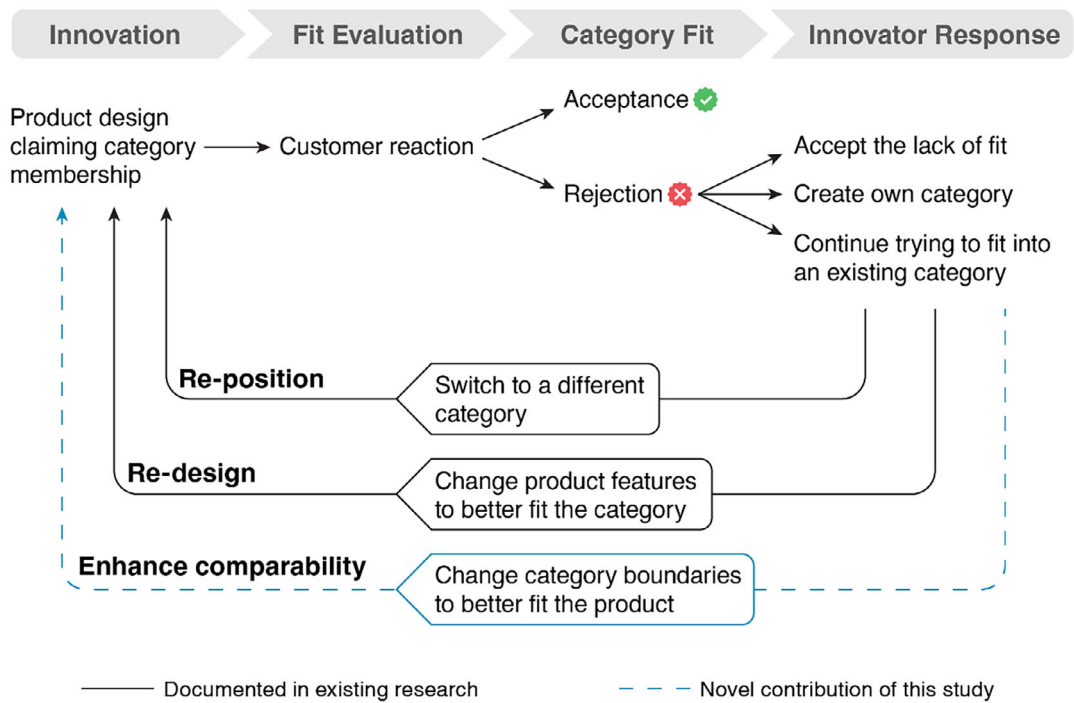


FIGURE 5 Positioning innovations in product categories.

thereby addressing their objections without further redesign or the creation of an entirely new category.

Figure 5 maps this paper's contribution to the literature on positioning innovations in product categories. The solid lines represent pathways discussed in existing research: innovators may (1) create a new category, or (2) reposition or redesign the product to fit into an existing category. The dashed line represents this paper's contribution: a third pathway in which innovators achieve category fit not by changing the product, but by changing the category itself.

The pathway put forward in this paper highlights a different understanding of the nature of categories compared to the two pathways documented in the existing literature. The finding that innovators outside the category can change the boundaries of an existing category suggests that rather than being rigidly defined constraints, categories are malleable structures that can be reshaped through sustained efforts at commensuration.

5 | DISCUSSION AND CONCLUSIONS

In positioning innovations within product categories, firms must choose between creating a new category or fitting their products into an existing one. While new category creation has been widely studied (e.g., Anthony et al., 2016; Nelson, 2025), the process of fitting innovations into existing categories has received less attention. This neglect stems, in part, from the assumption that such integration is straightforward, and that any challenges can be resolved through repositioning or product redesign (e.g., Zuckerman, 2000). According to this “category imperative” view, if efforts at redesign or repositioning fail to secure category membership, the



innovation faces penalties—reduced demand, worse market evaluations, and lower prices (e.g., Hsu et al., 2009).

This paper documents a complementary, yet distinct, pathway for fitting innovations into existing categories. Specifically, I show how, after decades of rejection, MBS issuers succeeded in changing the boundaries of the bond category to accommodate their products. Rather than conforming to the category through redesign or repositioning, MBS issuers added a new dimension—prepayment risk—to the existing category. This entailed transforming mortgage prepayments—a key difference between mortgages and bonds—into a type of risk and aligning it with call risk (the risk of early bond repayment). The fundamental contribution of this paper is to show that innovators can fit new products into existing categories by redefining the category's boundaries.

These findings contribute to the emerging literature on firm strategy and categories (e.g., Pontikes, 2018), the demand-based view of strategy (e.g., Adner, 2002; Adner & Zemsky, 2006; Priem, 2007), and the sociology literature on commensuration (e.g., Espeland & Stevens, 1998). They also highlight the value of an historical approach in strategy research (e.g., Argyres et al., 2020).

Recent work in categorization has called for greater attention to how firms strategically deploy categories (e.g., Pontikes, 2018)—including decisions about which categories to enter (e.g., Pontikes & Kim, 2017). Strategy scholars increasingly view categories not as fixed elements of the competitive landscape but as dynamic tools that firms can engage with to shape customer preferences (e.g., Anthony et al., 2016; Vinokurova, 2019). Category scholars have also shown that category meanings can evolve (e.g., Delmestri et al., 2020; Kennedy et al., 2010), and that category insiders can help bring about this change (e.g., Durand et al., 2007; Hsu & Grodal, 2015). This paper advances the research on the evolution of categories by showing how innovators outside the category can change the boundaries of an existing category to accommodate their products.

The finding is especially striking considering the conservatism of the target customers—bond investors—who “all like something that's been done before” (*Savings & Loan News*, Home mortgage-backed securities raise new capital for lenders, 1977, p. 22). This conservatism contrasts with other settings where products straddling categories can attract forward-thinking investors (e.g., Pontikes, 2012). MBS issuers addressed investor conservatism by examining the causes of early repayment in both MBS and bonds, and then aligning prepayment risk with call risk. Rather than merely redesigning MBS to resemble bonds, they changed the bond category boundaries to incorporate prepayment risk.

By emphasizing the role of changing category boundaries in innovation diffusion, this paper connects category theory with strategy and innovation research that highlights the importance of evaluation criteria (e.g., Christensen et al., 2008; Vinokurova & Kapoor, 2020). Strategy researchers have long argued that shaping evaluation criteria can be critical in the diffusion of innovation (Adner & Levinthal, 2008). This paper conceptualizes product categories as sets of evaluation criteria, offering a case study of how innovators can influence the audience's evaluative lens.

While much category scholarship has focused on prototype-based categories, an emerging body of research considers goal-based categories, where members meet the same need even if they differ in product category (e.g., Durand & Paoletta, 2013). For instance, an apple, yogurt, cup of coffee, or candy bar might all satisfy a customer looking for an afternoon snack. One way of assessing membership in a goal-based category is through substitution-in-use—can two products meet the same need? (e.g., Hamilton et al., 2014; Ratneshwar & Shocker, 1991;

Wansink, 1994). Although scholars have started to explore how markets function in goal-based categories (Glaser et al., 2020), we know less about how these interact with prototype-based categories. This paper shows how efforts to join a goal-based category (as a category substitute) can redefine the boundaries of a prototype-based category.

The demand-based view of strategy emphasizes how understanding customer preferences can be a source of competitive advantage (e.g., Adner, 2002; Adner & Zemsky, 2006; Priem, 2007; Schmidt, Priem & Zanella, 2024). Scholars have shown how such insights can guide strategic decisions like diversification (e.g., Mawdsley & Somaya, 2018; Ye et al., 2012). This paper explores how firms can leverage their understanding of customer perceptions of existing categories to facilitate innovation adoption—offering a bridge between demand-based strategy and firm category strategies (e.g., Pontikes & Kim, 2017). In so doing, the paper brings behaviorally realistic assumptions to the study of positioning in strategy (e.g., Adner, Csaszar, and Zemsky, 2014).

This paper also contributes to the literature on commensuration—an emerging stream of research in sociology that investigates how domains become comparable (e.g., Espeland & Stevens, 1998). Prior work in this stream has emphasized the role of quantification (turning qualitative assessments into numbers) in facilitating the commensuration process (e.g., Mennicken & Espeland, 2019). I show that quantification—turning prepayments into measurable risk—was necessary, but not sufficient for promoting commensuration of previously incomparable securities.

Equally important for aligning the newly quantified dimension to the comparison domain was the process of sensemaking. In the MBS case, sensemaking took the form of aligning the mechanisms behind call risk and prepayment risk (i.e., early repayment behavior by bond issuers and mortgage borrowers). This alignment enabled the mapping of both onto a single dimension of early repayment risk, thereby facilitating the inclusion of prepayment risk in the bond category. These findings point to a need for deeper investigation of sensemaking in commensuration research.

5.1 | Limitations, practitioner implications, and future directions

This paper is based on a single, in-depth case study—an extreme example of innovators altering a category's boundaries to achieve product acceptance. Future research is needed to investigate whether similar approaches could work in other contexts. Such research could shed light on which characteristics of the context: the social good nature of the product, the fragmentation of the market participants, the lack of a central regulating authority make it particularly ripe for category redefinition. Moreover, as an historical case study, the findings are bound by the time period in question. While it is possible that changes in technology or other societal trends may alter the way customers evaluate innovations, the cognitive and social processes underlying these changes are likely to be more enduring.

This paper offers three important lessons for management practitioners. First, innovators should consider a range of strategies for achieving product-category fit. Second, changing the boundaries of an existing category may be an attractive alternative to creating a new category or redesigning the product. Third, this paper underscores the value of historical analysis in strategic decision-making—while also warning against overgeneralizing from past experiences.

Future research could further develop the connections between the demand-based view of strategy and cognition by applying insights from the literature on managerial cognition and



strategic decision-making to the firm's customers (e.g., Csaszar, 2018; Csaszar & Levinthal, 2016; Gavetti & Levinthal, 2000; Gavetti & Menon, 2016). This line of inquiry might examine how strategic representations and cognitive limitations shape customer decision-making and how firms can leverage such representations and limitations to gain competitive advantage.

Another direction for future research is investigating the conditions under which firms can reshape evaluative criteria (e.g., the category test code). Prior research suggests that such conditions include evaluative uncertainty (i.e., the evaluators' lack of certainty as to what criteria they should be using) (Vinokurova & Kapoor, 2020). While category scholars have examined category change (e.g., Lo et al., 2020), further work is needed to understand what conditions promote category malleability.

Finally, prior work has posited that innovators can promote adoption by imitating existing products (e.g., Hargadon & Douglas, 2001). This paper suggests a complementary path: innovators can change how customers perceive category fit—not by altering the product, but by transforming the category itself.

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REFERENCES

- Adams, R. S., Stoddard, R. L., & Stempel, F. T. (1971). Investors view the GNMA certificate. *Mortgage Banker*, pp. 64–77.
- Adner, R. (2002). When are technologies disruptive? A demand-based view of the emergence of competition. *Strategic Management Journal*, 23(8), 667–688. <https://doi.org/10.1002/smj.246>.
- Adner, R., & Levinthal, D. (2008). Doing versus seeing: Acts of exploitation and perceptions of exploration. *Strategic Entrepreneurship Journal*, 2(1), 43–52.
- Adner, R., & Zemsky, P. (2006). A demand-based perspective on sustainable competitive advantage. *Strategic Management Journal*, 27(3), 215–239. <https://doi.org/10.1002/smj.513>.
- Adner, R., Csaszar, F. A., & Zemsky, P. B. (2014). Positioning on a multiattribute landscape. *Management Science*, 60(11), 2794–2815. <https://doi.org/10.1287/mnsc.2014.1978>.
- American Banker. (1985). Interview: Lewis S. Ranieri, Managing Director, Salomon Brothers, p. 22.
- Anderson, E. W., & Shugan, S. M. (1991). Repositioning for changing preferences: The case of beef versus poultry. *Journal of Consumer Research*, 18(2), 219–232.

- Anthony, C., Nelson, A. J., & Tripsas, M. (2016). "Who are you?... I really wanna know": Product meaning and competitive positioning in the nascent synthesizer industry. *Strategy Science*, 1(3), 163–183.
- Argyres, N. S., De Massis, A., Foss, N. J., Frattini, F., Jones, G., & Silverman, B. S. (2020). History-informed strategy research: The promise of history and historical research methods in advancing strategy scholarship. *Strategic Management Journal*, 41(3), 343–368.
- Boghossian, J., & David, R. J. (2021). Under the umbrella: Goal-derived category construction and product category nesting. *Administrative Science Quarterly*, 66(4), 1084–1129.
- Borowsky, M. (1993). Banks struggle to control mortgage prepayment risk. *Banking Management*, 69, 42–46.
- Business Week*. (1983, October 24). Making mortgages more appetizing for investors. pp. 170, 171.
- Carroll, M., & Lappen, A. A. (1994). Mortgage-backed mayhem. *Institutional Investor*, 28, 81–96.
- Carruthers, B. G., & Stinchcombe, A. L. (1999). The social structure of liquidity: Flexibility, markets, and states. *Theory and Society*, 28(3), 353–382.
- Cattani, G., Dunbar, R. L. M., & Shapira, Z. (2017). How commitment to craftsmanship leads to unique value: Steinway & Sons' differentiation strategy. *Strategy Science*, 2(1), 13–38.
- Christensen, C. M., Kaufman, S. P., & Shih, W. C. (2008). Innovation killers. *Harvard Business Review*, 86(1), 98–105.
- Connolly, J. J. (1977). Historical analysis of the GNMA market—Its yields and yield spread. *Mortgage Banker*, pp. 83–85.
- Conte, C., & Schellhardt, T. D. (1983, July 11). Big secondary market in mortgages smooths flow of housing funds. *Wall Street Journal*, 1–16.
- Csaszar, F. A. (2018). What makes a decision strategic? Strategic representations. *Strategy Science*, 3(4), 606–619.
- Csaszar, F. A., & Levinthal, D. A. (2016). Mental representation and the discovery of new strategies. *Strategic Management Journal*, 37(10), 2031–2049.
- David, R. J., & Lee, Y. (2022). The short history and long future of research on market categories. *Strategic Organization*, 20(4), 709–721.
- Davis, G. F., & Kim, S. (2015). Financialization of the economy. *Annual Review of Sociology*, 41(1), 203–221.
- Decker, S. (2013). The silence of the archives: Business history, post-colonialism and archival ethnography. *Management & Organizational History*, 8(2), 155–173.
- Delmestri, G., Wezel, F. C., Goodrick, E., & Washington, M. (2020). The hidden paths of category research: Climbing new heights and slippery slopes. *Organization Studies*, 41(7), 909–920.
- Draper, J. (1976). Ginnie Maes—More attractive than ever. *Pension World*, pp. 29–41.
- Durand, R., & Paoletta, L. (2013). Category stretching: Reorienting research on categories in strategy, entrepreneurship, and organization theory. *Journal of Management Studies*, 50(6), 1100–1123.
- Durand, R., Rao, H., & Monin, P. (2007). Code and conduct in French cuisine: Impact of code changes on external evaluations. *Strategic Management Journal*, 28(5), 455–472.
- Ennis, T. W. (1960, April 24). Mortgagees turn to pension funds: Bankers push efforts to persuade trustees to buy more property loans. *New York Times*, p. R1.
- Espeland, W. N., & Stevens, M. L. (1998). Commensuration as a social process. *Annual Review of Sociology*, 24(1), 313–343.
- Federal Home Loan Mortgage Corporation. (1983). Annual report.
- Federal Home Loan Mortgage Corporation. (1985). Annual report.
- Financial Publishing Company. (1972a). GNMA Serials, *Mortgage Banker*, p. 30.
- Financial Publishing Company. (1972b). Passing thru, *Mortgage Banker*, p. 13.
- Fink, L. D. (1996). The role of pension funds and other investors. In L. T. Kendall & M. J. Fishman (Eds.), *A primer on securitization* (pp. 117–127). MIT Press.
- Fishback, P. V., Rose, J., & Snowden, K. (2013). The patchwork mortgage market in the 1920s. In *Well worth saving: How the new Deal safeguarded home ownership* (pp. 9–19). University of Chicago Press.
- Ganis, D. (1974). All about the GNMA mortgage-backed securities market. *Real Estate Review*, 4, 55–65.
- Gastineau, G. L. (1992). *Dictionary of financial risk management*. Probus.
- Gastineau, G. L., & Kritzman, M. P. (1996). *Dictionary of financial risk management*. Frank J. Fabozzi Associates.
- Gavetti, G., & Levinthal, D. (2000). Looking forward and looking backward: Cognitive and experiential search. *Administrative Science Quarterly*, 45(1), 113–137.



- Gavetti, G., & Menon, A. (2016). Evolution cum agency: Toward a model of strategic foresight. *Strategy Science*, 1(3), 207–233. <https://doi.org/10.1287/stsc.2016.0018>.
- Glaser, V. L., Krikorian Atkinson, M., & Fiss, P. C. (2020). Goal-based categorization: Dynamic classification in the display advertising industry. *Organization Studies*, 41(7), 921–943.
- GNMA Mortgage-Backed Securities Dealers Association. (1978). *The Ginnie Mae manual*. Dow-Jones Irwin.
- Goldman, D. P. (1995). Sell callables, buy junk. *Forbes*, p. 256.
- Government National Mortgage Association. (1973). Annual report.
- Government National Mortgage Association. (1981). Annual report.
- Graham, M. B. W., & Shuldiner, A. T. (2001). *Corning and the craft of innovation*. Oxford University Press.
- Granqvist, N., Grodal, S., & Woolley, J. L. (2013). Hedging your bets: Explaining executives' market labeling strategies in nanotechnology. *Organization Science*, 24(2), 395–413.
- Gregan-Paxton, J., & Moreau, P. (2003). How do consumers transfer existing knowledge? A comparison of analogy and categorization effects. *Journal of Consumer Psychology*, 13(4), 422–430.
- Hamecs, R. (1984). CMOs and how they grew. *Mortgage Banking*, 44, 9–15.
- Hamilton, R. W., Thompson, D. V., Arens, Z. G., Blanchard, S. J., Häubl, G., Kannan, P. K., Khan, U., Lehmann, D. R., Meloy, M. G., Roese, N. J., & Thomas, M. (2014). Consumer substitution decisions: An integrative framework. *Marketing Letters*, 25, 305–317.
- Haney, R. L. (1978). An analysis of yield spreads between Ginnie Mae pass-throughs and Aaa corporate bonds. *Financial Management*, 7(1), 17–28.
- Hannan, M. T., Pólos, L., & Carroll, G. R. (2012). *Logics of organization theory: Audiences, codes, and ecologies*. Princeton University Press.
- Hargadon, A. B., & Douglas, Y. (2001). When innovations meet institutions: Edison and the design of the electric light. *Administrative Science Quarterly*, 46(3), 476–501.
- Hess, A. P., Jr., & Winn, W. J. (1962). *The value of the call privilege*. University of Pennsylvania.
- Homer, S. (1962). How to buy bonds. *Barron's*, 5, 13–17.
- Hsu, G. (2006). Jacks of all trades and masters of none: Audiences' reactions to spanning genres in feature film production. *Administrative Science Quarterly*, 51(3), 420–450.
- Hsu, G., & Grodal, S. (2015). Category taken-for-grantedness as a strategic opportunity: The case of light cigarettes, 1964 to 1993. *American Sociological Review*, 80(1), 28–62.
- Hsu, G., Hannan, M. T., & Koçak, Ö. (2009). Multiple category memberships in markets: An integrative theory and two empirical tests. *American Sociological Review*, 74(1), 150–169.
- Ingram, P., Rao, H., & Silverman, B. S. (2012). History in strategy research: What, why, and how? In S. J. Kahl, B. S. Silverman, & M. A. Cusumano (Eds.), *Advances in Strategic Management* (Vol. 29, pp. 241–273). Emerald Group Publishing Limited.
- Jones, C., Maoret, M., Massa, F. G., & Svejenova, S. (2012). Rebels with a cause: Formation, contestation, and expansion of the de novo category “modern architecture,” 1870–1975. *Organization Science*, 23(6), 1523–1545.
- Jones, O. H. (1971). “The wave of the future” revisited. *Mortgage Banker*.
- Jones, O., & Grebler, L. (1961). *The secondary mortgage market: Its purpose, performance, and potential*. Real Estate Research Program, Graduate School of Business Administration.
- Jourdan, J. P., Bureau, R., Rochais, C., & Dallemagne, P. (2020). Drug repositioning: A brief overview. *Journal of Pharmacy and Pharmacology*, 72(9), 1145–1151.
- Journal of Accountancy*. (1991, November). FASB issues ED on prepayment risk. p. 27.
- Kanner, S. B. (1982). Financing ideas: The secondary market for mortgages. *Real Estate Law Journal*, 10, 344–348.
- Kaufman, H. (1986). *Interest rates, the markets, and the new financial world*. Times Books.
- Kennedy, M. T. (2008). Getting counted: Markets, media, and reality. *American Sociological Review*, 73(2), 270–295.
- Kennedy, M. T., Lo, J., & Lounsbury, M. (2010). Category currency: A framework for analyzing the effects of meaning construction process. *Research in the Sociology of Organizations*, 31, 369–397.
- Khaire, M., & Wadhvani, R. D. (2010). Changing landscapes: The construction of meaning and value in a new market category—Modern Indian art. *Academy of Management Journal*, 53(6), 1281–1304.
- Kidd, P. E. (1970). GNMA mortgage-backed security: The wave of the future. *Mortgage Banker*, pp. 36–42.

- Kidd, P. E. (1971). One year old and going strong! *Mortgage Banker*, pp. 24–31, 38–40.
- Kingman, W. (1970). Mortgage-backed security. *Mortgage Banker*, pp. 66–68.
- Kirk, J. (1974). What's wrong with that nice girl Ginnie Mae? *Pension & Welfare News*, 110, 20–23.
- Kovács, B., Hsu, G., & Sharkey, A. (2024). The stickiness of category labels: Audience perception and evaluation of producer repositioning in creative markets. *Management Science*, 70(9), 6315–6335.
- Krippner, G. R. (2012). *Capitalizing on crisis: The political origins of the rise of finance*. Harvard University Press.
- Leibowitz, M. L., & Lechner, M. N. (1974). Measuring and managing the mortgage vs. bonds yield relationship. *Pension & Welfare News*, 10, 18–32, 37, 61.
- Lo, J. Y., Fiss, P. C., Rhee, E. Y., & Kennedy, M. T. (2020). Category viability: Balanced levels of coherence and distinctiveness. *Academy of Management Review*, 45(1), 85–108.
- Loehwing, D. A. (1970). Fat years and lien: Mortgage bankers seem to flourish in any kind of economy. *Barron's*, 50, 3.
- Lubinski, C., & Wadhvani, R. D. (2020). Geopolitical jockeying: Economic nationalism and multinational strategy in historical perspective. *Strategic Management Journal*, 41(3), 400–421.
- MacKenzie, D. (2011). The credit crisis as a problem in the sociology of knowledge. *American Journal of Sociology*, 116(6), 1778–1841.
- Mawdsley, J. K., & Somaya, D. (2018). Demand-side strategy, relational advantage, and partner-driven corporate scope: The case for client-led diversification. *Strategic Management Journal*, 39(7), 1834–1859.
- Meagher, J. P. (1983). A bond's encore. *Barron's*, 63, 84.
- Mennicken, A., & Espeland, W. N. (2019). What's new with numbers? Sociological approaches to the study of quantification. *Annual Review of Sociology*, 45, 223–245.
- Moffitt, D. (1978). Ginnie Mae pass-through certificates offer high yields plus safety for cautious savers. *Wall Street Journal*, 48.
- Moore, G. (2014). *Crossing the chasm* (3rd ed.). HarperCollins.
- Navis, C., & Glynn, M. A. (2010). How new market categories emerge: Temporal dynamics of legitimacy, identity, and entrepreneurship in satellite radio, 1990–2005. *Administrative Science Quarterly*, 55(3), 439–471.
- Navis, C., & Glynn, M. A. (2011). Legitimate distinctiveness and the entrepreneurial identity: Influence on investor judgments of new venture plausibility. *Academy of Management Review*, 36(3), 479–499.
- Nelson, A. J. (2025). “Even better than the real thing”? Electronic organs and the dilemma of product mimicry. *Strategic Management Journal*, 46(6), 1428–1452. <https://doi.org/10.1002/smj.3704>.
- Nevins, L. H. (1972). Mortgage-backed securities—A multifaceted investment tool. *Savings Bank Journal*, 53, 23–41.
- New York Times*. (1983, September 23). Mortgage agency to sell collateralized securities, p. D7.
- Pensions & Investments*. (1975, July 21). Fund managers shun mortgages out of habit, p. 3.
- Pensions & Investments*. (1977, November 21). Bright future seen for mortgage device, p. 10.
- Perry, J. R. (1979). Pass-through securities: Making the market in pension portfolios. *Pension World*, pp. 53–56.
- Peters, H. F. (1979). *Termination distributions of FHA insured residential mortgages*. (Ph.D. dissertation). University of Pennsylvania.
- Pontikes, E. G. (2012). Two sides of the same coin: How ambiguous classification affects multiple audiences' evaluations. *Administrative Science Quarterly*, 57(1), 81–118.
- Pontikes, E. G. (2018). Category strategy for firm advantage. *Strategy Science*, 3(4), 620–631.
- Pontikes, E. G., & Kim, R. (2017). Strategic categorization. *Research in the Sociology of Organizations*, 51, 71–111.
- Priem, R. L. (2007). A consumer perspective on value creation. *Academy of Management Review*, 32(1), 219–235.
- Quint, M. (1981, December 19). Your money: Pass-through securities. *New York Times*, p. 34.
- Ranieri, L. S. (1996). The origins of securitization, sources of its growth, and its future potential. In L. T. Kendall & M. J. Fishman (Eds.), *A primer on securitization* (pp. 31–44). MIT Press.
- Ratneshwar, S., & Shocker, A. D. (1991). Substitution in use and the role of usage context in product category structures. *Journal of Marketing Research*, 28(3), 281–295.
- Rhee, E. Y., Lo, J. Y., & Durand, R. (2025). Meeting expectations: Category legitimation and transmutation. *Organization Science*. <https://doi.org/10.1287/orsc.2023.17957>
- Rhee, E. Y., Lo, J. Y., Kennedy, M. T., & Fiss, P. C. (2017). Things that last? Category creation, imprinting, and durability. *Research in the Sociology of Organizations*, 51, 295–325.



- Rosa, J. A., Porac, J. F., Runser-Spanjol, J., & Saxon, M. S. (1999). Sociocognitive dynamics in a product market. *Journal of Marketing*, 63(4), 64–77.
- Rosen, K. (1975). The mortgage market—High time America's pension-funds moved in. *Pension World*, pp. 19–25, 67–68.
- Rowlinson, M., Hassard, J., & Decker, S. (2014). Research strategies for organizational history: A dialogue between historical theory and organization theory. *Academy of Management Review*, 39(3), 250–274.
- Savings & Loan News*. (1977, January). Home mortgage-backed securities raise new capital for lenders, pp. 18–22.
- Savings Institutions. (1983). FHLMC Three-tiered bond scores big, pp. 116–119.
- Schmidt, J., Priem, R., & Zanella, P. (2024). Customers, markets, and five archetypical value creation logics: A review of demand-side research in strategic management. *Journal of Management*, 50(6), 2309–2342. <https://doi.org/10.1177/01492063241227157>.
- Seamans, R., & Zhu, F. (2017). Repositioning and cost-cutting: The impact of competition on platform strategies. *Strategy Science*, 2(2), 83–99.
- Securities and Exchange Commission. (1995). Improving descriptions of risk by mutual funds and other investment companies. *Federal Register*, 60(64), 17172–17181.
- Senft, D. E. (1979). The “true yield” of a pass-through security. *Mortgage Banker*, pp. 15–19.
- Sloane, L. (1983). *New version of Ginnie Mae* (p. 30). New York Times.
- Snoddy, B. O., & Dickinson, R. C. (1971). Yes—Some problems... But... A giant step forward. *Mortgage Banker*, pp. 78–83.
- Sojacy, L. (1980). Mortgage securities outshine others. *Pensions & Investments*, pp. 24–25, 29.
- Star, M. G. (1992). Use of mortgage-backeds doubles. *Pensions & Investments*, pp. 3, 37.
- Struck, R. (1974). The average life of a single-family mortgage. *FHLBB Journal*, 7, 15–20.
- Urban, G. L., Hulland, J. S., & Weinberg, B. D. (1993). Premarket forecasting for new consumer durable goods: Modeling categorization, elimination, and consideration phenomena. *Journal of Marketing*, 57(2), 47–63.
- Vaara, E., & Lamber, J. A. (2016). Taking historical embeddedness seriously: Three historical approaches to advance strategy process and practice research. *Academy of Management Review*, 41(4), 633–657.
- Villas-Boas, J. M. (2018). A dynamic model of repositioning. *Marketing Science*, 37(2), 279–293.
- Vinokurova, N. (2018a). Failure to learn from failure: The 2008 mortgage crisis as a déjà vu of the mortgage meltdown of 1994. *Business History*, 61(6), 1005–1050. <https://doi.org/10.1080/00076791.2018.1440548>.
- Vinokurova, N. (2018b). How mortgage-backed securities became bonds: The emergence, evolution, and acceptance of mortgage-backed securities in the United States, 1960–1987. *Enterprise & Society*, 19(3), 610–660. <https://doi.org/10.1017/eso.2017.45>.
- Vinokurova, N. (2019). Reshaping demand landscapes: How firms change customer preferences to better fit their products. *Strategic Management Journal*, 40(13), 2107–2137. <https://doi.org/10.1002/smj.3074>.
- Vinokurova, N. (2020). A history of markets past: The role of institutional memory failure in financial crises. In R. Mir & A. L. Fayard (Eds.), *The Routledge companion to anthropology and business* (pp. 34–63). Routledge.
- Vinokurova, N., & Kapoor, R. (2020). Converting inventions into innovations in large firms: How inventors at Xerox navigated the innovation process to commercialize their ideas. *Strategic Management Journal*, 41(13), 2372–2399.
- Wadhwani, R. D. (2023). Critical hermeneutics: Deriving meaning from historical sources. In S. Decker, W. M. Foster, & E. Giovannoni (Eds.), *Handbook of historical methods for management* (pp. 218–231). Edward Elgar Publishing.
- Wadhwani, R. D., & Decker, S. (2017). Clio's toolkit: The practice of historical methods in organization studies. In R. Mir & S. Jain (Eds.), *The Routledge companion to qualitative research in organization studies* (pp. 113–127). Routledge.
- Wall Street Journal*. (1973, September 14). Government weighs S&L bond sale idea as a way to lessen the mortgage pinch, p. 2.
- Wall Street Journal*. (1979, April 20). Display Ad 103, p. 39.
- Wang, R. D., & Shaver, J. M. (2014). Competition-driven repositioning. *Strategic Management Journal*, 35(11), 1585–1604.
- Wansink, B. (1994). Advertising's impact on category substitution. *Journal of Marketing Research*, 31(4), 505–515.

- White, S. (1976). Everyone into the pool: Ginnie Mae passthroughs have made quite a splash. *Barron's*, 56, 11–14.
- Ye, G., Priem, R. L., & Alshwer, A. A. (2012). Achieving demand-side synergy from strategic diversification: How combining mundane assets can leverage consumer utilities. *Organization Science*, 23(1), 207–224.
- Zigas, D. (1983, November 15). Corporate issuers face competition, Analyst predicts. *The Bond Buyer*, p. 3
- Zuckerman, E. W. (1999). The categorical imperative: Securities analysts and the illegitimacy discount. *American Journal of Sociology*, 104(5), 1398–1438.
- Zuckerman, E. W. (2000). Focusing the corporate product: Securities analysts and de-diversification. *Administrative Science Quarterly*, 45(3), 591–619.

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